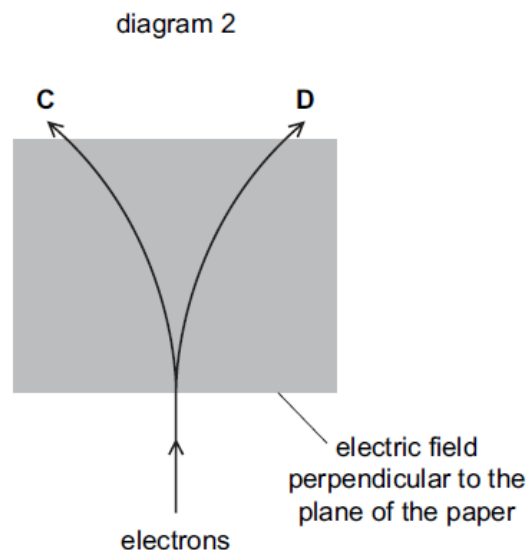
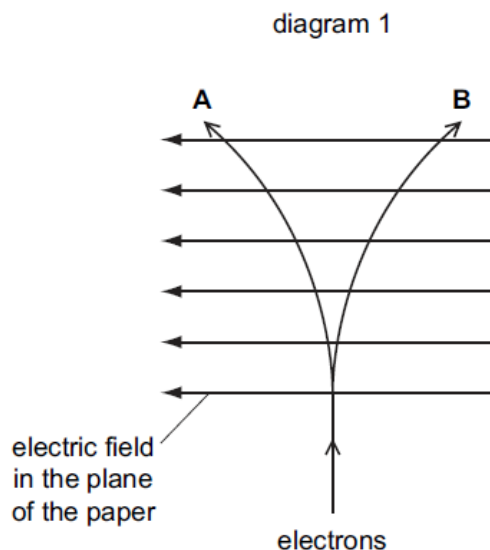


18 A beam of electrons is directed into an electric field and is deflected by it.

Diagram 1 represents an electric field in the plane of the paper. Diagram 2 represents an electric field directed perpendicular to the plane of the paper.

The lines A, B, C and D represent possible paths of the electron beam. All paths are in the plane of the paper.

Which line best represents the path of the electrons inside the field?



Answer: B

Electrons are negatively charged. In the left diagram, the field points to the left, so the force on electrons is to the right. An upward moving electron will curve towards the right → Path B.

For the diagram on the right, electric field is either in or out of the paper. This will cause electrons to deflect out of the plane of the paper, not along any of the shown paths which lie in the plane.

19 An isolated point charge is placed at point Q. The electric potential at point X is measured to be